Leonardo Guevara Atehortúa

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Systems Engineering student with 4 years of hands-on experience in software development through academic, personal, and applied projects. Skilled in Python, Java, TypeScript, SQL, backend development, data processing, full-stack applications, machine learning, analytics and AI-powered solutions. Experienced working collaboratively throughout software development processes applying Scrum practices. Among my projects, I participated in the development of FluxLab, a LIMS (Laboratory Information Management System) for environmental biotechnology laboratories, currently used in the biotechnology laboratory at the University of Medellin. My experience also includes working with databases, analyzing technical and business requirements, and contributing to the improvement and evolution of software solutions.

SKILLS

- **Languages**: Python, JavaScript, TypeScript, Bash, SQL.
- **Backend**: NestJS, Node.js, Django, REST APIs, JWT.
- **Frontend**: HTML, CSS, Angular, React.
- **Databases**: PostgreSQL, MySQL, SQLite, Supabase.
- **Data & Machine Learning:** Pandas, NumPy, Scikit-learn, TensorFlow, Power BI, Supervised Learning, Unsupervised Learning.
- **Cloud & DevOps**: Google Cloud Platform (GCP), GKE, Cloud SQL, Secret Manager, IAM, Workload Identity, Docker, Kubernetes, CI/CD.
- **AI & Agents:** LangChain, LangGraph, OpenAI SDK, MCP.
- **Other Tools**: VS Code, MQTT, Git, SonarQube, Jira, Kibana, Elasticsearch, Flutter, MQTT.

EXPERIENCE AND PROJECTS

FluxLab
LIMS web platform for managing biotechnology clients, projects, and samples

- Developed a full-stack solution for the Biotechnology Laboratory at the University of Medellín, using a NestJS backend and a React frontend to centralize clients, projects, samples, and reports into a unified operational platform.
- Implemented sample templates with dynamic fields, data types, and configurable ordering, ensuring consistency, traceability, and standardization of information.
- Developed dashboards with metrics, recent activity, and workflow statuses, as well as Excel-exportable reports with summaries by template and status to facilitate decision-making and result delivery.
- Implemented role-based access control and JWT authentication integration through Supabase.
- **Demo**: fluxlab-front.vercel.app/

FocusHub
Web application for managing tasks, events, focus sessions, and productivity statistics.

- Designed and developed the complete productivity workflow using a NestJS backend and an Angular frontend, including task management with statuses, deadlines, duplication, and advanced filters.

- Implemented a productivity module with a calendar (month/week/day) and focus techniques such as Pomodoro and custom sessions, enabling event creation, task visualization by date, and timer state persistence.
- Implemented JWT authentication and endpoint protection, using TypeORM with PostgreSQL to ensure secure access and data persistence.
- Developed statistical dashboards with charts on focus hours, completed tasks, and productivity trends, as well as a daily progress panel.

  Demo: focushub-eight.vercel.app/

Medical Documents API (CI/CD project)
Medical Documents API & Multi-Cloud Orchestration Platform designed and operated a distributed, event-driven system of 3 independently deployed REST APIs across GCP, AWS, and OCI, coordinated by a custom asynchronous orchestrator, with Kubernetes orchestration and full observability.

- Developed an exposed RESTful API using Node.js and Fastify, implementing CRUD operations and the non-standard HTTP QUERY method for filtered data retrieval.
- Built and published versioned Docker images to Google Artifact Registry for Kubernetes and Cloud Run deployments.
- Built an event-driven orchestration service using Google Cloud Pub/Sub for asynchronous cross-cloud processing, message acknowledgment, automatic retries, and distributed trace propagation across GCP, AWS, and OCI.
- Deployed the API on GKE with 2 replicas, HPA (2–5 pods), startup/readiness/liveness probes, resource requests/limits, ConfigMaps, Secrets, Service Accounts, and LoadBalancer networking.
- Integrated PostgreSQL on Google Cloud SQL with Cloud SQL Auth Proxy, securely connecting the Kubernetes workload to managed database infrastructure.
- Orchestrator deployed to Cloud Run via Cloud Build.
- Implemented distributed observability with OpenTelemetry and Grafana Cloud, centralizing traces, metrics, and logs across GCP, AWS, and OCI, with trace-based correlation and latency/error monitoring.
- Applied a structured Git/GitHub workflow using GitMoji, feature branches, Pull Requests, and isolated environment configurations to maintain controlled and traceable releases.
- GitHub repository: github.com/LeonardoG2005/MedicalDevops

Ergonominador
IoT monitoring system (Django, MQTT, ESP32, SQLite, Chart.js)

- Developed an IoT system integrating ESP32 sensors with a Django backend through MQTT protocols.
- Implemented real-time data ingestion and persistence with SQLite.
- Developed a REST API and dashboard with Chart.js for ergonomic monitoring.
- Designed an alert system based on posture and environmental conditions.
- Demo: ergonominador.vercel.app/

Personal Portfolio
Responsive personal portfolio website

- Designed and developed a responsive personal portfolio website using semantic HTML5, CSS3, and vanilla JavaScript to present my projects, technical skills, experience, and professional profile.
- Implemented a modern UI with CSS variables, Grid/Flexbox layouts, animated gradients, smooth scrolling, and responsive breakpoints for mobile, tablet, and desktop devices.
- Developed interactive project sections with dynamic modals, live demo links, project documentation, and detailed technology and architecture information.

- **GitHub repository**: github.com/LeonardoG2005/Portfolio

Bank Marketing

Binary classification model

- Developed a binary classification model to predict whether bank customers would subscribe to a term deposit based on historical marketing campaign data.
- Performed exploratory data analysis, preprocessing, feature encoding, and model evaluation using stratified 10-fold cross-validation and F1-score.
- Compared Logistic Regression and SVM models, including hyperparameter tuning and decision-threshold optimization to improve the balance between Precision and Recall.
- **GitHub repository**: github.com/LeonardoG2005/Bank-Marketing-Database

EDUCATION

Universidad de Medellín — **Medellin, Colombia**

Systems Engineering (9th semester). — *Jul 2022 - Present*

IELTS Academic Certification

C1 English Proficiency

- C1 English proficiency validated through IELTS Academic.

Machine Learning Specialization

DeepLearning.AI

- Training in supervised and unsupervised learning, neural networks, and model implementation.
- Application of concepts in real-world projects involving data pipelines, analytics, and intelligent systems.

Data Engineer in Python Career Track

DataCamp

- Hands-on training in data engineering, including data processing and data manipulation.
- Experience using tools such as Pandas and SQL for data transformation and analysis.
- Hands-on experience in building data workflows and managing structured datasets.

LANGUAGES

- English: C1 (IELTS Certification).
- Spanish: Native.